

Customer Shopping Behavior Analysis

1. Project Overview

The project analyzes customer shopping behavior by using data from multiple tables and 4390 rows. The goal is to uncover insights into spending patterns, customer segments, product preferences, and reason behind return products to guide strategic business decisions.

2. Dataset Summary

-- tables: 7

- (Customers, order_items, orders, products, returns, reviews)

-- columns: 39

(customer_id, customer_name, city, state, gender, age_group, singup_date, acquisition_channel, order_item_id, order_id, product_id, quantity, unit_price, discount_pct, item_total, customer_id, order_date, order_states, payment_method, sales_channel, gross_amount, discount_amount, shipping_fee, final_amount, delivered_date, product_name, category, concern, skin_type, key_ingredient, size, mrp, cost_price, stock_qty, launch_date, return_id, return_date, return_reason, refund_status, review_id, rating, review_date).

-- rows: 4390

4. Data Analysis using SQL

We performed structured analysis in MS SQL to answer key business questions:

1. **most discount used by age group:** Compared all the age group by total used discount

Results			Messages		
	age_group	total_discount			
1	18-24	33685.799908638			
2	45+	11929.1000070572			
3	35-44	16627.4999885559			
4	25-34	62428.6002206802			

2. **Age group by most purchase:** Identified customers age group who purchases most

	age_group	total_orders
1	25-34	632
2	18-24	313
3	35-44	182
4	45+	123

3. **Total orders by genders:** Identified the total number of orders by genders

Results		Messages
	gender	total_order
1	Male	441
2	Female	433
3	Other	376

4. **Top 10 city by most purchases:** Identified top 10 city where we sale most

	top_city	orders
1	Nagpur	92
2	Indore	89
3	Lucknow	89
4	Surat	87
5	Kochi	68
6	Gurugram	68
7	Pune	67
8	Bhopal	65
9	Bengaluru	62
10	Mumbai	62

5. **Top sale channels:** Top sale channels where we sale most from.

	num_of_orders	sales_channel
1	698	Website
2	444	Mobile App
3	108	Marketplace

6. **Top 5 products by rating:** Found products with the highest average review rating.

	product_name	avg_rating
1	Aqualogica Lightweight Sunscreen	4
2	Body Lotion with Niacinamide	4
3	Bond Repair Shampoo	4
4	Glycolic Acid Face Cleanser	4
5	Hyaluronic Acid 2% Serum	4

7. **Identified repeated buyers:** Identified customers who are repeated buyers (only those who buy more than 5 time)

Results		Messages
	customer_name	num_of_order
1	Aadhya Mishra	13
2	Kiara Das	13
3	Krishna Reddy	12
4	Meera Reddy	12
5	Aditya Shama	12
6	Sai Kumar	12
7	Yash Trivedi	12
8	Aarav Iyer	11
9	Priya Agarwal	11
10	Priya Trivedi	11
11	Kiara Patel	11
12	Karan Mishra	11
13	Neha Shama	10
14	Ira Iyer	10
15	Rohan Das	10
16	Reyansh Gupta	10
17	Reyansh Rao	9
18	Vivaan Nair	9
19	Arjun Gupta	9
20	Arjun Patel	9

8. **Top reasons for returns:** found the top reasons behind the returns.

Results		Messages
	return_reason	num_of_returns
1	Skin irritation	28
2	Late delivery	16
3	Damaged packaging	13
4	Wrong item received	12
5	Changed mind	10

9. **Top customers acquisition channels:** Identified top channels where we acquired most of customers from.

Results Messages		
	acquisition_channel	num_of_customers
1	Google Search	148
2	Instagram	145
3	YouTube	78
4	Website Direct	70
5	Referral	59

10. **High-spending discounted users:** customers who used discount but still spend more than average purchase amount

100 %  No issues found

 Results

 Messages

	customer_id	final_amount
1	C0001	997.5
2	C0005	2047.25
3	C0006	1278.40002441406
4	C0007	923.049987792969
5	C0008	1088.25
6	C0009	958.400024414063
7	C0010	1429.90002441406
8	C0013	1906.09997558594
9	C0015	998.25
10	C0016	1118.15002441406
11	C0018	1258.19995117188
12	C0020	1998.80004882813
13	C0020	1706.30004882813
14	C0020	1362.19995117188
15	C0022	1497.15002441406
16	C0022	1347.30004882813
17	C0023	1188.30004882813

5. Dashboard in Power BI

At last, we built an interactive dashboard in **Power BI** to present insights visually.

